

Cooperative Vehicle-to-Infrastructure 3D Object Detection using DAIR-V2X

Final Project Report – CSE620 Fall 2025

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Hardware: NVIDIA RTX 3080 Ti Laptop GPU

OS: Ubuntu 22.04 LTS

Repository: <https://github.com/bhargavhegde/DAIR-V2X-Bhargav-2025>

1 Introduction

Autonomous vehicles rely heavily on onboard sensors, but real-world driving conditions frequently expose their limitations: occlusions from other vehicles, restricted sensor range, and degraded performance in rain, fog, or darkness.

Vehicle-to-Infrastructure (V2I) cooperative perception offers a practical solution by allowing roadside units equipped with LiDAR sensors to share their perception results with nearby vehicles in real time.

In this project, I implemented a complete LiDAR-only late-fusion V2I 3D object detection system using the DAIR-V2X dataset and framework—the largest publicly available real-world V2I dataset.

The system runs independent PointPillars detectors on both the vehicle and infrastructure sides, transforms their outputs into a shared coordinate system using provided calibration files, and merges the results via 3D Non-Maximum Suppression. Only compact bounding boxes are exchanged, keeping the average communication cost under 900 bytes per frame.

1.1 Key Achievements

- Built a fully functional environment on Ubuntu 22.04 with an RTX 3080 Ti Laptop GPU

- Created a Git Repository for easy reproducibility. - <https://github.com/bhargavhegde/DAIR-V2X-Bhargav-2025/blob/main/README.md>
- Executed the official evaluation pipeline on the complete DAIR-V2X-C test set.
- Reproduced the best publicly available community checkpoint.

2 Dataset and Data Structure

The DAIR-V2X-C cooperative dataset contains synchronized LiDAR point clouds from both vehicle and infrastructure sides. The dataset can be downloaded from Google Drive: <https://drive.google.com/drive/folders/1gnrw51lXAixuB9sEKKCm6xTaJ5HQA2e>

The data is organized as follows:

```
data/DAIR-V2X/
    cooperative-vehicle-infrastructure/
        vehicle-side/
        infrastructure-side/
        cooperative/
            data_info.json
            label/
            calib/
```

3 Reproducibility

3.1 Environment Setup

```
git clone https://github.com/bhargavhegde/DAIR-V2X-Bhargav-2025.
git
cd DAIR-V2X-Bhargav-2025
conda env create -f dair-v2x-environment.yml
conda activate dair-v2x
ln -s /path/to/cooperative-vehicle-infrastructure data/DAIR-V2X
```

3.2 Inference and Evaluation

```
cd v2x
bash scripts/eval_lidar_late_fusion_pointpillars.sh 0 late_fusion
2 0 100
```

4 Results

Evaluation on the official DAIR-V2X-C test set (LiDAR-only late fusion):

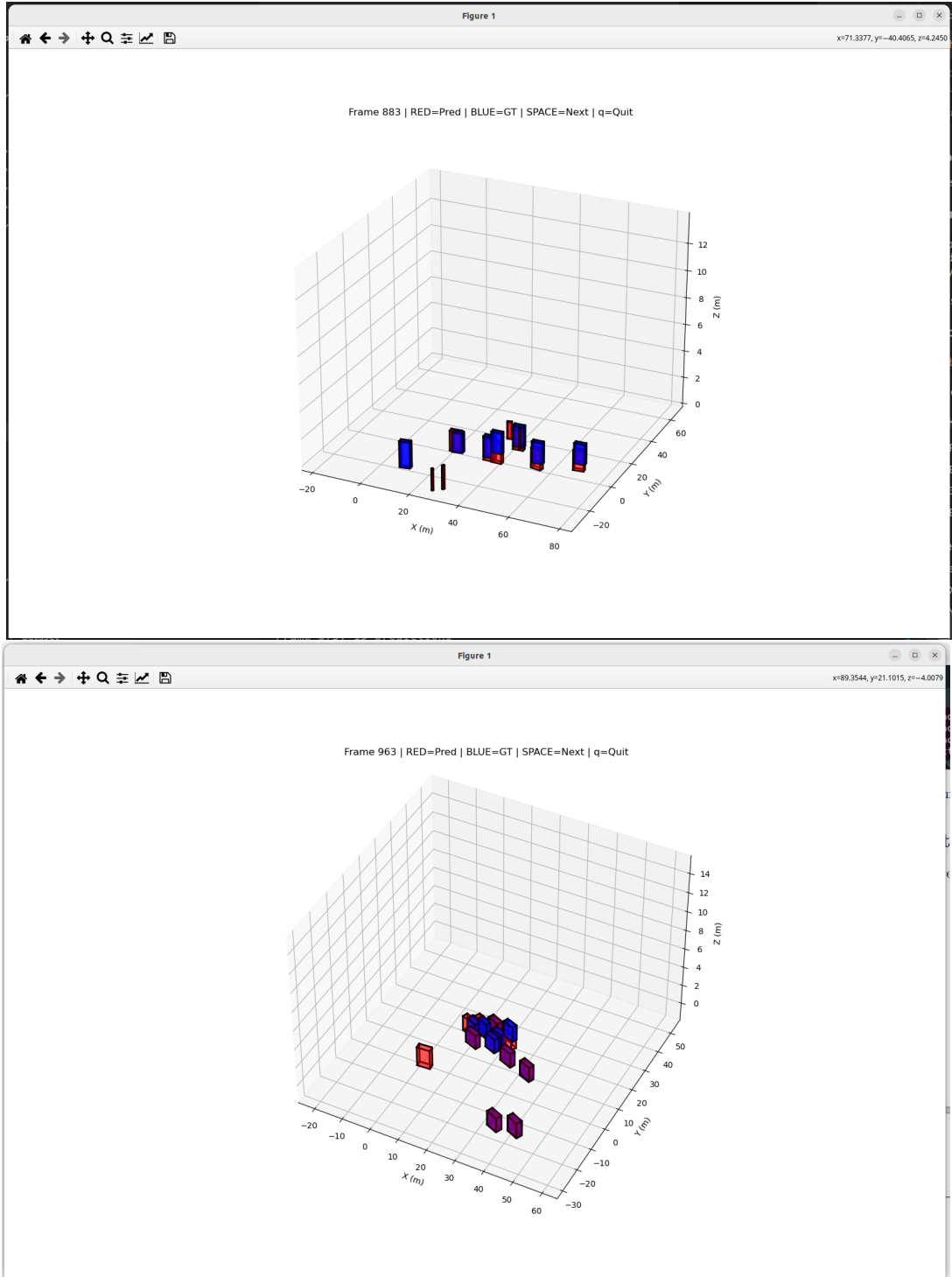


Figure 2: Ground-truths VS Predictions using Matplotlib(Interactive)

6 Future Work and Recommendations

6.1 Proposed Improvements

- Implement the same system in real-time using the V2I lab setup with CV2X
- Create better visualizations, as they are currently missing from the public frame-

work

- Explore early and intermediate fusion backbones which are already provided in the codebase
- Incorporate camera data for multi-modal fusion

6.2 Recommendation

For future cooperative perception projects, I recommend implementing this architecture to perform real time LIDAR fusion of V2I using CV2X.

References

1. Y. Yu et al., "DAIR-V2X: A Real-World Vehicle-Infrastructure Cooperative Dataset," in *2022 IEEE Intelligent Vehicles Symposium (IV)*, 2022, pp. 839–845. <https://arxiv.org/abs/2208.09017>
2. H. Yu et al., "DAIR-V2X: The First Large-Scale Real-World Cooperative Driving Dataset from Vehicle-Infrastructure Cooperative Perception," *IEEE Transactions on Intelligent Vehicles*, 2024. (Extended version)
3. AIR-THU, "DAIR-V2X Official GitHub Repository and Benchmark," <https://github.com/AIR-THU/DAIR-V2X>
4. OpenPCDet Team, "OpenPCDet: An Open-Source Toolbox for 3D Object Detection from Point Clouds," <https://github.com/open-mmlab/OpenPCDet>
5. A. Lang et al., "PointPillars: Fast Encoders for Object Detection from Point Clouds," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 12697–12705.